# $\mathbf{P}_{12}$ for complex phases: a proof

*Files:* `scratchpad/ideaB/`. *Numerical sanity checks of every step:* `check_chain.py` *($P_{12}$),* `check_chain_t2.py` *(Target 2).*

**Setting.** $1 \le s \le \frac{3}{2}$, $r = 1/s \in [\frac{2}{3}, 1]$, $\theta = 3^{1-s}$, $c = c_{12}(s) = 1 - (\frac{2}{3})^{s-1}$. Atoms $\beta_i \in \mathbb{C} \setminus \{0\}$, $\xi_i, \eta_i \in \mathbb{T}$ $(i = 1, \dots, m)$, $A = \sum \beta_i$, $B = \sum \beta_i \xi_i$, $C = \sum \beta_i \eta_i$, $D = \sum \beta_i \xi_i \eta_i$, $Q = \sum |\beta_i|^r$.

**Theorem 1** ($\mathrm{P}_{12}$)**.** *If $B = C = 0$ then $|A| - c\,|D| \le \theta Q^s$. Equality holds for the aligned pair $(\xi, \eta) = (1, 1), (-1, -1)$, $\beta_1 = \beta_2$.*

**Notation for the proof.** Write $\beta_i = \rho_i p_i$ with $\rho_i > 0$, $p_i \in \mathbb{T}$; $L = \sum_i \rho_i$ $(\ge |A|)$. Put

$$x_i = p_i \xi_i, \qquad y_i = p_i^2 \xi_i \eta_i, \qquad D' = \sum_i \rho_i y_i, \qquad \delta' = |D'|/L \in [0, 1].$$

Then $B = \sum \rho_i x_i$, $C = \sum \rho_i \bar{x}_i y_i$, $A = \sum \rho_i p_i$, $D = \sum_i \rho_i \bar{p}_i y_i$. Finally

$$\Phi(\delta) = \tfrac{1}{2}\big[(1+\delta)^r (3+\delta)^{1-r} + (1-\delta)^r (3-\delta)^{1-r}\big], \qquad \varphi(\delta) = 3^{r-1}\Phi(\delta), \qquad \text{so } \theta\Phi(\delta)^s = \varphi(\delta)^s.$$

**Lemma 2** (twisted leverage bound)**.** *If $B = C = 0$ then $Q \ge L^r\, \Phi(\delta')$.*

*Proof.* Let $u_i = (1, x_i, y_i) \in \mathbb{T}^3$ and $H = \sum_i \rho_i u_i u_i^*$. Its diagonal is $L$; its entries below the diagonal are $\sum \rho_i x_i = B = 0$, $\sum \rho_i y_i \bar{x}_i = C = 0$ and $\sum \rho_i y_i = D'$. So $H = \begin{pmatrix} L & 0 & \bar{D}' \\ 0 & L & 0 \\ D' & 0 & L \end{pmatrix} \succeq 0$. For each $i$, $H \succeq \rho_i u_i u_i^*$ and $u_i \in \operatorname{ran} H$, hence $\rho_i w_i \le 1$ with $w_i = u_i^* H^+ u_i$ (leverage). Since $r \le 1$, $\rho_i^r = \rho_i \rho_i^{r-1} \ge \rho_i w_i^{1-r}$.

If $\delta' < 1$, $H$ is invertible and, with $\omega = D'/|D'|$ (any $\omega \in \mathbb{T}$ if $D' = 0$) and $X_i = \operatorname{Re}(\bar{\omega} y_i) \in [-1, 1]$,

$$w_i = \frac{1}{L} + \frac{2L - 2\operatorname{Re}(\bar{D}' y_i)}{L^2 - |D'|^2} = \frac{3 - \delta'^2 - 2\delta' X_i}{L(1 - \delta'^2)}, \qquad \sum_i \rho_i X_i = \operatorname{Re}(\bar{\omega} D') = L\delta'.$$

The function $g(X) = \big((3 - \delta'^2 - 2\delta' X)/(1 - \delta'^2)\big)^{1-r}$ is concave on $[-1, 1]$, so $g(X) \ge \frac{1+X}{2} g(1) + \frac{1-X}{2} g(-1)$, with $g(1) = (\frac{3+\delta'}{1+\delta'})^{1-r}$, $g(-1) = (\frac{3-\delta'}{1-\delta'})^{1-r}$. Hence

$$Q \ge L^r \sum_i \tfrac{\rho_i}{L} g(X_i) \ge L^r \Big[\tfrac{1+\delta'}{2} g(1) + \tfrac{1-\delta'}{2} g(-1)\Big] = L^r \Phi(\delta').$$

If $\delta' = 1$ then $y_i = \omega$ for all $i$, $\operatorname{ran} H = \operatorname{span}\{(1, 0, \omega), (0, 1, 0)\}$ with eigenvalues $2L, L$, so $w_i = \frac{2}{2L} + \frac{1}{L} = \frac{2}{L}$ and $Q \ge L^r 2^{1-r} = L^r \Phi(1)$. □

**Lemma 3** (Jensen)**.** *For every $C \ge 0$: $|A| - C|D| \le L\sqrt{1 + C^2 - 2C\delta'}$.*

*Proof.* Let $\alpha = A/|A|$ (or 1), $\omega$ as above, $P_i = \bar{\alpha} p_i$, $Y_i = \bar{\omega} y_i$. Then $\sum \rho_i P_i = |A|$, $\sum \rho_i Y_i = L\delta'$, and $D = \bar{\alpha}\omega \sum \rho_i \bar{P}_i Y_i$, so $|D| \ge \operatorname{Re} \sum \rho_i \bar{P}_i Y_i$. Therefore

$$|A| - C|D| \le \sum_i \rho_i \operatorname{Re}\big(\bar{P}_i(1 - C Y_i)\big) \le \sum_i \rho_i |1 - C Y_i| \le L\Big(\tfrac{1}{L} \sum_i \rho_i |1 - C Y_i|^2\Big)^{1/2} = L\sqrt{1 + C^2 - 2C\delta'}.$$

□

**Lemma 4** (one variable)**.** *For $1 \le s \le \frac{3}{2}$, $0 \le \delta \le 1$ and $C = \min(\delta, c)$:* $\varphi(\delta)^s \ge \sqrt{1 + C^2 - 2C\delta}$.

*Proof.* For $s = 1$ both sides equal 1; let $s > 1$. Write $\varphi(\delta) = \frac{1}{2}(1+\delta)^r(1+\frac{\delta}{3})^{1-r} + \frac{1}{2}(1-\delta)^r(1-\frac{\delta}{3})^{1-r}$.

*(a) $\delta \le c$.* Here the claim is $\varphi \ge (1-\delta^2)^{r/2}$. With $\frac{1}{2}(e^u + e^v) = e^{(u+v)/2}\cosh\frac{u-v}{2}$,

$$\varphi(\delta) = (1-\delta^2)^{r/2}\,(1-\tfrac{\delta^2}{9})^{\frac{1-r}{2}}\cosh Y, \qquad Y = r\,\operatorname{artanh}\delta + (1-r)\operatorname{artanh}\tfrac{\delta}{3} \;\ge \tfrac{1+2r}{3}\delta \ge \tfrac{7}{9}\delta.$$

Using $\cosh Y \ge 1 + Y^2/2$ and $(1-x)^a \ge 1 - ax/(1-x)$ ($a = \frac{1-r}{2} \le \frac{1}{6}$, $x = \delta^2/9 \le \frac{1}{9}$): $(1 - \frac{\delta^2}{9})^{\frac{1-r}{2}}\cosh Y \ge (1 - \frac{\delta^2}{48})(1 + \frac{49}{162}\delta^2) \ge 1$. (This holds for all $\delta \in [0,1]$.)

*(b) $c \le \delta \le 1$.* Here the claim is $\varphi^{2s} \ge 1 + c^2 - 2c\delta = (1-c)^2 + 2c(1-\delta)$. Put $\pi_\pm = \frac{1\pm\delta}{2}$, $a_+ = \frac{3+\delta}{3(1+\delta)}$, $a_- = \frac{3-\delta}{3(1-\delta)}$; then $\varphi = \pi_+ a_+^{1-r} + \pi_- a_-^{1-r}$, and Jensen (exp convex) gives $\varphi \ge e^{(1-r)\kappa(\delta)}$, where

$$\kappa(\delta) = \pi_+\log a_+ + \pi_-\log a_- = -\sum_{k\ge1} a_k\delta^{2k}, \qquad a_k = \frac{1-(4k+1)9^{-k}}{2k(2k-1)} \in \Big(0, \frac{1}{2k(2k-1)}\Big),\ a_1 = \tfrac{2}{9}$$

(expand $\pi_+\log(1+\frac{\delta}{3}) + \pi_-\log(1-\frac{\delta}{3})$ and $\pi_+\log(1+\delta) + \pi_-\log(1-\delta)$). Facts:

(K1) $\kappa \le 0$, $|\kappa(\delta)| \le \frac{2}{9}\delta^2 + \frac{\delta^4}{12(1-\delta^2)}$, $\kappa(1) = \log\frac{2}{3}$.

(K2) $e^\kappa$ is concave on $[0,1]$: $\kappa' = -\operatorname{artanh}\delta + \operatorname{artanh}\frac{\delta}{3} + \frac{2\delta}{9-\delta^2} \in [-\operatorname{artanh}\delta, 0]$ and $\kappa'' = -\frac{1}{1-\delta^2} + \frac{45-\delta^2}{(9-\delta^2)^2} \le -\cosh^2 y + \frac{11}{16}$ with $y = \operatorname{artanh}\delta$, so $\kappa'' + \kappa'^2 \le y^2 + \frac{11}{16} - (1 + y^2 + \frac{y^4}{4}) \le -\frac{5}{16}$.

(K3) Hence $\frac{2}{3} \le e^\kappa \le 1$, i.e. $\lambda := \kappa/\log\frac{2}{3} \in [0,1]$.

With $t = s-1$ and $x = (\frac{2}{3})^t = 1 - c \in [x_0, 1)$, $x_0 = \sqrt{2/3}$: $\varphi^{2s} \ge e^{2t\kappa} = x^{2\lambda}$. So it suffices that $F(\xi) := \xi^{2\lambda} - \xi^2 - 2(1-\xi)(1-\delta) \ge 0$ at $\xi = x$. Now $F(1) = 0$ and $F$ is concave on $[x_0, 1]$: $F'' = 2\lambda(2\lambda-1)\xi^{2\lambda-2} - 2 \le \max\big(-2,\ 2\lambda(2\lambda-1)(\frac{3}{2})^{1-\lambda} - 2\big) \le 0$, because $\lambda \mapsto 2\lambda(2\lambda-1)(\frac{3}{2})^{1-\lambda}$ increases on $[\frac{1}{2}, 1]$ to 2. The constraint $c \le \delta$ is $x \ge 1-\delta$, so $x \in [\xi_L, 1]$ with $\xi_L = \max(x_0, 1-\delta)$, and it suffices that $F(\xi_L) \ge 0$:

- $\delta \ge 1 - x_0$ ($\xi_L = x_0$): $F(x_0) = e^{\kappa(\delta)} - \frac{2}{3} - 2(1-x_0)(1-\delta)$. This is concave in $\delta$ (K2) minus linear, $= 0$ at $\delta = 1$, and at $\delta = 1 - x_0 = 0.18350\ldots$ it is $\ge e^{-0.0075808} - (2x_0 - \frac{2}{3}) = 0.99244 - 0.96633 > 0$ by (K1).
- $\delta < 1 - x_0$ ($\xi_L = 1-\delta$): $F(1-\delta) = (1-\delta)^{2\lambda} - (1-\delta^2) \ge -2\lambda|\log(1-\delta)| + \delta^2$. By (K1) $\lambda \le 0.556\,\delta^2$ and $|\log(1-\delta)| \le 1.235\,\delta$ for $\delta \le 0.19$, so $F \ge \delta^2(1 - 1.38\,\delta) > 0$. □

*Proof of Theorem 1.* Let $C = \min(\delta', c) \le c$. By Lemmas 3, 4, 2,

$$|A| - c|D| \le |A| - C|D| \le L\sqrt{1 + C^2 - 2C\delta'} \le L\varphi(\delta')^s = \theta\big(L^r\Phi(\delta')\big)^s \le \theta Q^s.$$

For the pair, $L = 1 = |A| = |D|$, $Q^s = 2^{s-1}$ and $|A| - c|D| = (\frac{2}{3})^{s-1} = \theta Q^s$. □

*Remark* 5 (what the proof gives beyond $\mathrm{P}_{12}$). (i) For all $C \ge 0$ (with $B = C_{\mathrm{sum}} = 0$): $|A| - C|D| \le L\sqrt{1 + C^2 - 2C\delta'}$ and $\theta Q^s \ge L\varphi(\delta')^s$; the two meet only at the IF block ($\delta' = 0$, $L = |A|$) and at the pair ($\delta' = 1$). (ii) Near the IF block this gives bonus $\le (\frac{2(1-r)}{9} + o(1))|D|^2/L$ (sharp local constant $\frac{1-r}{8+r}$). (iii) For $\beta_i > 0$ Lemma 2 alone (with $p_i = 1$) plus the one-variable inequality is the whole proof; it is a Rényi-entropy statement for a probability measure on $\mathbb{T}^2$ with $\mathbb{E}\xi = \mathbb{E}\eta = 0$.

## TARGET 2: the sharp pointwise constant $c_1 = 1 - 3^{1-s}$

**Theorem 6.** *For all atoms (no constraint) and $1 \le s \le \frac{3}{2}$: $|A| \le \theta Q^s + (1-\theta)\max(|B|,|C|,|D|)$. Equality for one atom.*

Keep the notation $\rho_i, p_i, L, x_i, y_i, D'$, and put $c_1 = 1-\theta$, $M = \max(|B|,|C|,|D|)$, $b = \sqrt{(|B|^2+|C|^2)/2}/L$, $y = |D'|/L$. Again $u_i = (1, x_i, y_i)$, $H = \sum \rho_i u_i u_i^* = L\hat{H}$; now $\hat{H}$ is a correlation matrix with $\hat{H}_{10} = B/L$, $\hat{H}_{21} = C/L$, $\hat{H}_{20} = D'/L$; eigenvalues $\nu_1, \nu_2, \nu_3 \ge 0$, $\sum \nu_k = 3$.

*Step A (cost lemma for $H$).* The paper's Sylvester/Hadamard step gives $\det(I + tH) \le \prod_i (1 + 3t\rho_i)$; integrating log against $t^{-1-r}dt$ gives $\sum_k (L\nu_k)^r \le 3^r Q$, i.e. $\theta Q^s \ge L\left(\frac{1}{3}\sum_k \nu_k^r\right)^s$.

*Step B (Jensen, weights $\nu_k/3$).* $\frac{1}{3}\sum \nu_k^r = \sum_k \frac{\nu_k}{3}\nu_k^{r-1} \ge \exp\big((r-1)\sum_k \frac{\nu_k}{3}\log\nu_k\big)$, so $\left(\frac{1}{3}\sum \nu_k^r\right)^s \ge \theta^\lambda$ with $\lambda = \frac{1}{3\log 3}\sum_k \nu_k \log\nu_k \in [0,1]$.

*Step C (spectral reduction).* $\nu_k = 1 + \mu_k$ with $\mu^3 - P\mu - q = 0$, $P = 2b^2 + y^2$, $q = 2\,\mathrm{Re}(\hat{H}_{10}\hat{H}_{21}\overline{\hat{H}_{20}}) \ge -2b^2 y$, and $q \ge P - 1$ (because $\nu_{\min} \ge 0$). Let $\Lambda(P,q) = \frac{1}{3\log 3}\sum h(\mu_k)$, $h(\mu) = (1+\mu)\log(1+\mu)$. For a monic cubic, $\sum_k g(\mu_k)/p'(\mu_k)$ is the divided difference $g[\mu_1,\mu_2,\mu_3] = g''(\xi)/2$; hence $\partial_q\Lambda \propto h'''(\xi)/2 < 0$ and $\partial_P\Lambda \propto (\mu h')''(\xi)/2 = \frac{1}{2}\big(\frac{1}{1+\xi} + \frac{1}{(1+\xi)^2}\big) > 0$. Therefore

$$\lambda = \Lambda(P,q) \le \tilde{\lambda}(b,y) := \Lambda\big(2b^2 + y^2,\ \max(-2b^2 y,\ 2b^2 + y^2 - 1)\big),$$

and $\tilde{\lambda}$ is increasing in $b$ and in $y$ (on the branch $q = -2b^2y$ by the two signs; on the branch $q = P-1$ the spectrum is $\{0, \frac{3\pm\sqrt{4P-3}}{2}\}$, which spreads as $P$ grows).

*Step D (upper bound for the left side).* For $\lambda' \in [0,1]$, $0 \le \kappa \le 1-\lambda'$ and $C = c_1\kappa$: since $c_1 M \ge c_1\lambda' L b + C|D|$, Lemma 3 (which does not use $B = C = 0$) gives $|A| - c_1 M \le L\big[\sqrt{1 + C^2 - 2Cy} - c_1\lambda' b\big]$.

*Step E (removing $s$).* Put $x = \theta \in [x_0, 1]$, $x_0 = 3^{-1/2}$. For fixed $(b, y, \lambda', \kappa)$, $G(x) = x^{\tilde{\lambda}} - \sqrt{1 + (1-x)^2\kappa^2 - 2(1-x)\kappa y} + (1-x)\lambda' b$ is concave on $[x_0, 1]$ ($x^{\tilde{\lambda}}$ concave, the square root is the norm of an affine function of $x$) and $G(1) = 0$. So $G(x_0) \ge 0$ implies $G(\theta) \ge 0$ for every $s \in [1, \frac{3}{2}]$.

*Step F (one $s$-free inequality on $[0,1]^2$).* For every $(b, y) \in [0,1]^2$ there are admissible $(\lambda', \kappa)$ with $G(x_0) \ge 0$, i.e.

$$\text{(S)} \qquad x_0^{\tilde{\lambda}(b,y)} \ \ge\ \sqrt{1 + c_0^2\kappa^2 - 2c_0\kappa y} - c_0\lambda' b, \qquad c_0 = 1 - x_0.$$

(F0) $[0, \frac{c_0}{2}] \times [0, c_0]$: $t\log t \le (t-1) + (t-1)^2$ for $t \ge 0$ gives $\sum \nu\log\nu \le 2P$, so $x_0^{\tilde{\lambda}} \ge e^{-P/3} \ge 1 - \frac{P}{3}$. If $c_0 b \ge y^2/2$ take $(\lambda', \kappa) = (1, 0)$ and use $\frac{P}{3} \le \frac{2b^2}{3} + \frac{2c_0 b}{3} \le c_0 b$; otherwise take $(0, y/c_0)$, right side $\sqrt{1-y^2} \le 1 - \frac{y^2}{2}$, and $b < \frac{y^2}{2c_0} \le \frac{y}{2}$ gives $\frac{P}{3} \le \frac{y^2}{2}$.

(F1) $[0.99, 1] \times [0.985, 1]$: here $q = P - 1$, the spectrum is $\{0, 3-\varepsilon, \varepsilon\}$ with $\varepsilon(3-\varepsilon) = \eta = 3 - P \in [2e_1 + e_2,\ 0.0696]$ ($e_1 = 1-b$, $e_2 = 1-y$), so $\varepsilon \le 0.02338$. Then $x_0^{\tilde{\lambda}} \ge x_0 + \frac{x_0}{6}\varepsilon\big[\log\frac{3}{\varepsilon} + 1 - \frac{\varepsilon}{6-2\varepsilon}\big] \ge x_0 + 0.5630\,\varepsilon$, while $(1,0)$ or $(0,1)$ gives right side $\le x_0 + c_0\min(e_1, e_2/x_0) \le x_0 + \frac{c_0\eta}{2+x_0} \le x_0 + 0.4920\,\varepsilon$.

(F2) The rest of $[0,1]^2$: certified by interval arithmetic (`t2_certify.py`, mpmath `iv`, 30 digits). By the monotonicity of $\tilde{\lambda}$ and of the right side, a box $[b_0, b_1] \times [y_0, y_1]$ is certified by $x_0^{\tilde{\lambda}(b_1, y_1)} \ge \mathrm{RHS}(b_0, y_0)$; roots of the cubic by the trigonometric formula in interval arithmetic. Result: 1479 boxes, no failure, minimal certified margin $2.47 \cdot 10^{-4}$.

*Conclusion.* $\theta Q^s \ge L\theta^\lambda \ge L\theta^{\tilde{\lambda}(b,y)} \ge L\big[\sqrt{1 + C^2 - 2Cy} - c_1\lambda' b\big] \ge |A| - c_1 M$. □

*Remark* 7 (a global quadratic bonus bound from the same chain). Steps A and C with the exact spectrum, and Lemma 3, give for all atoms $|A|-\theta Q^s \le L\big[\min_{C\ge 0}(C\,d+\sqrt{1+C^2-2Cy})-S(b,y)\big]$, $d=|D|/L$, $S=$ the worst-phase value of $(\frac{1}{3}\sum \nu_k^r)^s$. Maximising the bracket divided by $2b^2+d^2$ (grid and local search, not certified) gives $|A|-\theta Q^s \le K\,(|B|^2+|C|^2+|D|^2)/|A|$ with $K=0.282$ at $s=\frac{3}{2}$ (maximiser $b\approx 0.425$, $y\approx 0.64$, $d\approx 0.45$), against the numerically sharp 0.275.